

ABBREVIATIONS

ACCU	AIR COOLED CONDENSING UNIT
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AI	ANALOG INPUT
AO	ANALOG OUTPUT
APD	AIR PRESSURE DROP
AS	AIR SEPERATOR
B	BOILER
BB	BASEBOARD
BC	BOOSTER COIL
BDD	BACKDRAFT DAMPER
BFG	BELOW FINISHED GRADE
BFP	BACKFLOW PREVENTER
BHP	BRAKE HORSE POWER
BOD	BOTTOM OF DUCT
BOP	BOTTOM OF PIPE
BT	BATH TUB
BTU	BRITISH THERMAL UNITS
BTUH	BRITISH THERMAL UNITS PER HOUR
C	CONVECTOR
CA	COMPRESSED AIR
CB	CATCH BASIN
CC	COILING COIL
CCF	100 CUBIC FEET
CD	CEILING DIFFUSER
CF	CUBIC FEET
CFH	CUBIC FEET PER HOUR
CFM	CUBIC FEET PER MINUTE
CFP	CLEAN OUT FERRULE AND PLUG
CH	CHILLER
CI	CAST IRON
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CO	CLEAN OUT
CO	CARBON MONOXIDE
COND	CONDENSATE
CO2	CARBON DIOXIDE
CT	COOLING TOWER
CU	CONDENSING UNIT
CU FT	CUBIC FOOT
CUH	CABINET UNIT HEATER
CW	COLD WATER
CWR	CONDENSER WATER RETURN
CWS	CONDENSER WATER SUPPLY
D	DIFFUSER
DB	DRY BULB
DEG F	DEGREE FAHRENHEIT
DI	DRINKING FOUNTAIN
DIA	DIGITAL INPUT
DN	DOWN
DO	DIGITAL OUTPUT
DVV	DRAIN, WASTE AND VENT
EA	EACH
EAT	ENTERING AIR TEMPERATURE
EC	ELECTRICAL CONTRACTOR
EF	EXHAUST FAN
EG	EXHAUST AIR GRILLE
ELEV	ELEVATION
ER	EXHAUST AIR REGISTER
EUH	ELECTRIC UNIT HEATER
EW	ELECTRIC WATER COOLER
EWV	ELECTRIC WATER HEATER
EWT	ENTERING WATER TEMPERATURE
EX	EXISTING
EXH	EXHAUST
F	FURNACE
FAI	FRESH AIR INTAKE
FBO	FURNISHED BY OTHERS
FC	FAN COIL
FD	FLOOR DRAIN
FD	FIRE DAMPER
FIN	FIN TUBE RADIATION
FLG	FLANGE
FP	FIRE PROTECTION
FPM	FEET PER MINUTE
FT	FOOT OR FEET
F&T	FLOAT AND THERMOSTATIC TRAP
G	GAS (NATURAL)
GAL	GALLON
GC	GENERAL CONTRACTOR
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GT	GREASE TRAP
H	HYDROGEN
HB	HOSE BIBB
HC	HEATING COIL
HCO	HORIZONTAL CLEANOUT
HP	HORSE POWER
HTR	HEATER
HVAC	HEATING, VENTILATING & AIR COND
HW	HOT WATER
HWC	HOT WATER RECIRCULATING
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
HX	HEAT EXCHANGER
IE	INVERT ELEVATION
IF	INCH OR INCHES
IN	IRON PIPE
IP	INVERT
IWV	INSTANTNEOUS WATER HEATER

JAN	JANITOR
JC	JANITOR'S CLOSET
JS	JANITOR'S SINK
KW	KILOWATT
LAT	LEAVING AIR TEMPERATURE
LAV	LAVATORY
LBS	POUNDS
LHWR	LOW TEMPERATURE HOT WATER
RETURN	LOW TEMPERATURE HOT WATER
LHWS	LOW TEMPERATURE HOT WATER
SUPPLY	LOW TEMPERATURE HOT WATER
LQ	LIQUID (REFRIGERATION)
LV	LOUVER
LP	PROPANE GAS (LIQUID)
LWT	LEAVING WATER TEMPERATURE
M	METER
MAU	MAKE-UP AIR HANDLING UNIT
MAX	MAXIMUM
MB	MOP BASIN
MBH	BRITISH THERMAL UNITS (1000)
MBTUH	BRITISH THERMAL UNITS (1000)
MC	MECHANICAL CONTRACTOR
MCA	MINIMUM CIRCUIT AMPACITY
MD	MOTORIZED DAMPER
MECH	MECHANICAL
MIN	MINIMUM
MOCP	MAXIMUM OVER CURRENT
PROTECTION	
N2O	NITROUS OXIDE
NC	NORMALLY CLOSED
NIC	NOT INCLUDED OR NOT IN CONTRACT
NIT	NITROGEN
NO	NORMALLY OPEN
OA	OUTDOOR AIR
OAI	OUTDOOR AIR INTAKE
OC	ON CENTER
OD	OVERFLOW DRAIN
ODP	OPEN DRIP PROOF
OXY	OXYGEN
P	PUMP
PC	PLUMBING CONTRACTOR
PD	PNEUMATIC OPERATED DAMPER
PV	POST INDICATING VALVE
PRV	PRESSURE REDUCING VALVE
PT	PRESSURE TEMPERATURE PLUG
PTAC	PACKAGED TERMINAL AIR COND. UNIT
PVAC	PROCESS VACUUM
R	REGISTER
RA	RETURN AIR
RD	ROOF DRAIN
RET	RETURN
RF	RETURN FAN
RG	RETURN AIR GRILLE
RPZ	REDUCED PRESSURE ZONE BACKFLOW PREVENTER
RR	RETURN REGISTER
RTU	ROOF TOP UNIT
S	SINK
SA	SUPPLY AIR
SAD	SUPPLY AIR DIFFUSER
SAF	SUPPLY AIR FAN
SAN	SANITARY SEWER
SD	SMOKE DETECTOR
SG	SUPPLY AIR GRILLE
SH	SHOWER HEAD
SHC	STEAM HEATING COIL
SR	SUPPLY REGISTER
SS	STAINLESS STEEL
ST	STORM
STM	STEAM
SUCT	SUCTION (REFRIGERATION)
TCC	TEMPERATURE CONTROL CONTRACTOR
TCD	TEMPERATURE CONTROL DAMPER
TCP	TEMPERATURE CONTROL PANEL
TD	TRANSFER DUCT
TG	TRANSFER AIR GRILLE
TT	THERMOSTATIC TRAP
TYP	TYPICAL
UG	UNDERGROUND
UH	UNIT HEATER
UV	UNIT VENTILATOR
V	VENT
VAC	VACUUM
VAV	VARIABLE AIR VOLUME BOX
VD	VOLUME DAMPER
VFD	VARIABLE FREQUENCY DRIVE
VS	VENT STACK (SANITARY)
VTR	VENT THRU ROOF (SANITARY)
VUV	VERTICAL UNIT VENTILATOR
W	WASTE
WB	WET BULB
WC	WATER CLOSET
W/O	WITHOUT
WH	WATER HEATER
WSHP	WATER SOURCE HEAT PUMP
NOTE:	NOT ALL ABBREVIATIONS AND/OR SYMBOLS ARE USED IN THIS SET OF DOCUMENTS.

SYMBOL LEGEND

	SUPPLY AIR DIFFUSER
	LINEAR DIFFUSER
	RETURN OR EXHAUST AIR GRILLE
	EXHAUST FAN - ROOF
	FRESH AIR INTAKE HOOD - ROOF
	RELIEF AIR HOOD - ROOF
	SUPPLY REGISTER
	RETURN OR EXHAUST REGISTER
	SQUARE TO ROUND TRANSITION
	VOLUME CONTROL DAMPER
	FIRE / SMOKE DAMPER (RATING DETERMINED BY WALL TYPE)
	MOTORIZED DAMPER
	DUCT SMOKE DETECTOR
	THERMOSTAT - 60° A.F.F.
	HUMIDISTAT - 60° A.F.F.
	AIR FLOW
	SUPPLY AIR DUCT UP
	SUPPLY AIR DUCT DOWN
	RETURN AIR DUCT UP
	RETURN AIR DUCT DOWN
	EXHAUST AIR DUCT UP
	EXHAUST AIR DUCT DOWN
	BALL VALVE
	BUTTERFLY VALVE
	CIRCUIT SETTER
	CHECK VALVE
	GATE VALVE
	GLOBE VALVE
	2-WAY CONTROL VALVE
	3-WAY CONTROL VALVE
	PRESSURE RELIEF VALVE
	BALANCE VALVE
	INLINE PUMP
	STRAINER
	CAP
	PIPING 90
	PIPING "T"
	METER
	CONNECTION TO EXISTING

PLUMBING PIPING

	CW	DOMESTIC COLD WATER - EXISTING
	- - - - - CW - - - - -	DOMESTIC COLD WATER - DEMO
	CW	DOMESTIC COLD WATER - NEW
	HW	DOMESTIC HOT WATER - EXISTING
	- - - - - HW - - - - -	DOMESTIC HOT WATER - DEMO
	HW	DOMESTIC HOT WATER - NEW
	HWC	DOMESTIC HOT WATER RECIRC - EXISTING
	- - - - - HWC - - - - -	DOMESTIC HOT WATER RECIRC DEMO
	HWC	DOMESTIC HOT WATER RECIRC - NEW
	SAN	SANITARY - EXISTING
	- - - - - SAN - - - - -	SANITARY - DEMO
	SAN	SANITARY - NEW
	V	VENT - EXISTING
	- - - - - V - - - - -	VENT - DEMO
	V	VENT - NEW
	ST	STORM - EXISTING
	- - - - - ST - - - - -	STORM - DEMO
	ST	STORM - NEW
	OD	OVERFLOW STORM - EXISTING
	- - - - - OD - - - - -	OVERFLOW STORM - DEMO
	OD	OVERFLOW STORM - NEW
	G	GAS - EXISTING
	- - - - - G - - - - -	GAS - DEMO
	G	GAS - NEW

DUCTWORK

	24x14 SA	SUPPLY AIR DUCT - EXISTING
	24x14 SA	SUPPLY AIR DUCT - DEMO
	24x14 SA	SUPPLY AIR DUCT - NEW
	24x14 RA	RETURN AIR DUCT - EXISTING
	24x14 RA	RETURN AIR DUCT - DEMO
	24x14 RA	RETURN AIR DUCT - NEW
	24x14 OA	OUTSIDE AIR DUCT - EXISTING
	24x14 OA	OUTSIDE AIR DUCT - DEMO
	24x14 OA	OUTSIDE AIR DUCT - NEW
	24x14 EA	EXHAUST AIR DUCT - EXISTING
	24x14 EA	EXHAUST AIR DUCT - DEMO
	24x14 EA	EXHAUST AIR DUCT - NEW

MECHANICAL PIPING

	HWS	HYDRONIC HOT WATER SUPPLY - EXISTING
	- - - - - HWS - - - - -	HYDRONIC HOT WATER SUPPLY - DEMO
	HWS	HYDRONIC HOT WATER SUPPLY - NEW
	HWR	HYDRONIC HOT WATER RETURN - EXISTING
	- - - - - HWR - - - - -	HYDRONIC HOT WATER RETURN - DEMO
	HWR	HYDRONIC HOT WATER RETURN - NEW
	FP	FIRE PROTECTION - EXISTING
	- - - - - FP - - - - -	FIRE PROTECTION - DEMO
	FP	FIRE PROTECTION - NEW
	CDS	CONDENSOR WATER SUPPLY
	- - - - - CDS - - - - -	CONDENSOR WATER SUPPLY - DEMO
	CDS	CONDENSOR WATER SUPPLY - EXIST
	CDR	CONDENSOR WATER RETURN
	- - - - - CDR - - - - -	CONDENSOR WATER RETURN - DEMO
	CDR	CONDENSOR WATER RETURN - EXIST
	CHWS	CHILLED WATER SUPPLY
	- - - - - CHWS - - - - -	CHILLED WATER SUPPLY - DEMO
	CHWS	CHILLED WATER SUPPLY - EXIST
	CHWR	CHILLED WATER RETURN
	- - - - - CHWR - - - - -	CHILLED WATER RETURN - DEMO
	CHWR	CHILLED WATER RETURN - EXIST
	AIR	COMPRESSED AIR - EXISTING
	- - - - - AIR - - - - -	COMPRESSED AIR - DEMO
	AIR	COMPRESSED AIR - NEW
	STM	STEAM - EXISTING
	- - - - - STM - - - - -	STEAM - DEMO
	STM	STEAM - NEW
	C	CONDENSATE - EXISTING
	- - - - - C - - - - -	CONDENSATE - DEMO
	C	CONDENSATE - NEW

NOTES

MECHANICAL/PLUMBING AND FIRE PROTECTION GENERAL NOTES

- THESE DRAWINGS ARE DIAGRAMMATIC IN CHARACTER AND DO NOT NECESSARILY INDICATE EVERY OFFSET, VALVE, FITTING, ETC. REQUIRED. CONTRACTOR IS RESPONSIBLE FOR FIELD ROUTING ALL DUCTWORK AND PIPING AT NO ADDITIONAL COST.
- THE CONTRACTOR SHALL PROVIDE ALL EQUIPMENT, MATERIALS AND OPERATIONS AND PERFORM ALL LABOR REQUIRED FOR INSTALLATIONS AS INDICATED THE DRAWINGS, IN THE SPECIFICATIONS AND AS REQUIRED BY LOCAL, STATE AND FEDERAL CODES, AND AS MAY BE REASONABLY IMPLIED TO ACCOMPLISH COMPLETE MECHANICAL, PLUMBING AND FIRE PROTECTION SYSTEMS.
- ALL ELECTRICAL DISCONNECTS REQUIRED PER NEC CODE SHALL BE FURNISHED AND INSTALLED BY THE ELECTRICAL CONTRACTOR UNLESS NOTED OTHERWISE.
- CONTRACTOR SHALL PROVIDE PRODUCTS AS SPECIFIED ON THE DRAWINGS AND SPECIFICATIONS, HOWEVER, WHERE THE WORDS "EQUAL TO" ARE USED, ADDITIONAL PRODUCTS MAY BE SUBMITTED AS PROPOSED SUBSTITUTIONS, BUT REQUIRE APPROVAL FROM ARCHITECT/ENGINEER.
- DESIGN DRAWINGS SHOW GENERAL ARRANGEMENT AND EXTENT OF WORK. THE DRAWINGS ARE DIAGRAMMATIC AND MAY NOT NECESSARILY BE DRAWN TO SCALE FOR PURPOSE OF CLARITY AND LEGIBILITY. IT IS INTENDED THAT ALL ITEMS BE LOCATED SYMMETRICALLY WITH ARCHITECTURAL ELEMENTS WHERE FEASIBLE AND BE INSTALLED TO AVOID OBSTRUCTIONS AND PRESERVE HEADROOM. CONTRACTOR SHALL REVIEW PLANS OF OTHER TRADES WITH HIS OWN WORK TO AVOID CONFLICTS AND INTERFERENCES. CONTRACTOR MUST MAKE USE OF ALL SOURCES OF INFORMATION INCLUDING DRAWINGS OF EQUIPMENT FURNISHED BY OTHERS. FAILURE TO REVIEW WORKING SPACES OR CHECK DIMENSIONS IN QUESTION SHALL NOT WARRANT CONFLICTS.
- DRAWINGS AND SPECIFICATIONS ARE INTENDED TO BE COMPLIMENTARY. WHERE DRAWINGS AND SPECIFICATIONS CONFLICT EACH OTHER, IT IS THE CONTRACTORS RESPONSIBILITY TO GET CLARIFICATION FROM THE ARCHITECT/ENGINEER PRIOR TO BIDDING. FAILURE TO GET CLARIFICATION SHALL NOT RESULT IN ADDITIONAL COST AND THE MORE STRINGENT SHALL BE USED AS INTENDED BASIS FOR BIDDING.
- PLANS AND SPECIFICATIONS ARE INTENDED TO CONFORM TO GOVERNING CODES AND STANDARDS. IF NON-CONFORMITIES ARE DISCOVERED WHILE BIDDING, OR PERFORMING THE WORK IMPLIED, BRING THE SAME TO THE ATTENTION OF THE ARCHITECT/ENGINEER FOR CLARIFICATION IN WRITING PRIOR TO SUBMITTING BID OR PROCEEDING WITH WORK. NON-CONFORMITIES OF CODE COMPLIANCE WORK INSTALLED AND CORRECTIONS REQUIRED WITHOUT CONSULTATION AND WRITTEN RESPONSE OF SAME BY THE ENGINEER WILL BE THE CONTRACTORS FINANCIAL RESPONSIBILITY.
- MOTORS SHALL BE PROVIDED BY MECHANICAL CONTRACTOR AS REQUIRED BY THE EQUIPMENT FURNISHED BY THE MECHANICAL CONTRACTOR. MOTORS TO BE SUITABLE FOR LOAD, DUTY, VOLTAGE, FREQUENCY, HAZARD, SERVICE AND LOCATION INTENDED. SINGLE PHASE MOTORS MUST HAVE INTEGRAL THERMAL OVERLOAD PROTECTION IN ADDITION TO THAT PROVIDED IN MOTOR CONTROLLERS. MOTORS TO CONFORM IN DESIGN AND PERFORMANCE TO THE MOTOR STANDARDS OF NEMA. MOTORS RATED FOR CONTINUOUS DUTY UNDER FULL LOAD WITH A MAXIMUM TEMPERATURE RISE OF 105 DEG F FOR OPEN, 125 DEG F FOR DRIP PROOF AND 130 DEG F FOR EXPLOSION PROOF AND TOTALLY ENCLOSED TYPES. SUPPLY MOTORS WITH BELT DRIVES WITH ADJUSTABLE BASES, REMOVABLE BELT GUARDS AND VARIABLE PITCH DRIVE PULLEY SELECTED SO THAT MIDPOINT OF VARIABLE RANGE OF PULLEY WILL DRIVE EQUIPMENT AT RATED SPEED. MOTORS 1 HP AND LARGER SHALL BE THREE PHASE (UNLESS OTHERWISE LISTED).
- MOTOR CONTROLLERS SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR FOR MOTORS FURNISHED BY THE MECHANICAL CONTRACTOR. MOTOR CONTROLLERS SHALL BE OF SIZES AND TYPES AS NEEDED TO MEET THE OPERATIONAL CONDITIONS AS REQUIRED BY THE SEQUENCE OF OPERATION. ELECTRICAL CONTRACTOR TO FURNISH AND INSTALL THE POWER CIRCUIT, LOCAL DISCONNECT AND CONNECTION TO MOTOR TERMINALS. MECHANICAL CONTRACTOR TO MOUNT MOTOR CONTROLLERS AND CONTROL COMPONENTS AND WIRE AND MAKE ALL FINAL CONTROL CONNECTIONS BETWEEN DEVICES.
- THE DRAWINGS INDICATE KNOWN UTILITY AND DRAINAGE LINES IN ACCORDANCE WITH THE INFORMATION FURNISHED TO THE ENGINEER. RESPONSIBILITY FOR LOCATING, UNCOVERING, DISPOSING OR MAINTAINING ALL EXISTING UTILITY LINES TO REST SOLELY WITH THE CONTRACTOR. VERIFY LOCATIONS AND DEPTHS OF SERVICE CONNECTION POINTS BEFORE PROCEEDING WITH CONSTRUCTION.
- CONTRACTOR SHALL CHECK EXISTING PREMISES BEFORE SUBMISSION OF BIDS TO CHECK ALL CONDITIONS WHICH MAY EFFECT THE PERFORMANCE OF THE WORK INVOLVED. NO ALLOWANCES OR EXTRA PAYMENT WILL BE MADE DUE TO CONTRACTOR'S FAILURE TO EXAMINE SITE AND FULLY DISCERN WORKING CONDITIONS.
- MECHANICAL CONTRACTOR SHALL RECEIVE, PROPERLY HOUSE, HANDLE, HOIST, AND DELIVER TO PROPER LOCATION EQUIPMENT AND OTHER MATERIALS REQUIRED FOR THIS CONTRACT.
- THE CONTRACTOR SHALL OBTAIN PERMITS, ARRANGE FOR INSPECTIONS, AND PAY FEES AND EXPENSES IN CONNECTION THEREWITH, AS A PART OF THE WORK REQUIRING SUCH PERMITS. EVERY EFFORT IS MADE TO DESCRIBE THE WORK REQUIREMENTS IN CONFORMITY WITH APPLICABLE CODES.
- THE CONTRACTOR SHALL REVIEW ANY ALTERNATES OF OTHER TRADES, AND PRICE THEIR BID TO ACCOUNT FOR ITEMS AFFECTING HIS WORK.
- PIPING PASSING THROUGH CORRIDORS, TUNNELS, CHASES, ETC. SHALL BE CONSIDERED FOR PROPER DRAINAGE. CONSULT WITH THE OTHER CONTRACTORS AND AVOID CONFLICT WITH LOCATION OF PIPING. ORDER OF PRIORITY FOR ALL PIPING AND CONDUITS TO BE INSTALLED SHALL BE AS FOLLOWS WITH THE HIGHEST PRIORITY LISTED FIRST.
 - PLUMBING DRAIN LINES
 - CONDENSATE LINES
 - DUCTWORK
 - FIRE PROTECTION
 - HOT AND COLD WATER PIPING
 - ELECTRICAL CONDUIT
- WORK ASSIGNMENTS INFERRED BY THE DRAWINGS AND NOTES INCLUDED IN THESE PROJECT DOCUMENTS ARE INFORMATIONAL ONLY AND ARE NOT INTENDED TO RELIEVE THE BIDDING CONTRACTOR OF HIS OBLIGATION TO THE OWNER TO PROVIDE A COMPLETE AND COORDINATED PROJECT. COMPREHENSIVE SUBCONTRACTOR COORDINATION AND FINAL WORK ASSIGNMENTS TO SUBCONTRACTORS ARE THE SOLE RESPONSIBILITY OF THE BIDDING CONTRACTOR.
- ALL DUCTS SERVING SUPPLY, RETURN AND EXHAUST TERMINALS SHALL BE PROVIDED WITH BALANCE DAMPERS. FOR CLARITY, ALL DAMPERS MAY NOT BE SHOWN ON PLANS.
- PROVIDE DIELECTRIC UNIONS/CONNECTIONS AT ALL JUNCTIONS OF DISSIMILAR METALS.



www.upea.com

IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES
BESSEMER, MI

DATE:

02/07/2024

ISSUED FOR:

BIDSET

IRON BELLE APARTMENTS
BUCKNELL ENTERPRISES

PROJECT NO: B280-03239

DESIGNED BY: SP

DRAWN BY: SP

CHECKED: CP

APPROVED: GP

MECHANICAL AND
PLUMBING
ABBREVIATIONS
AND SYMBOLS

MP001

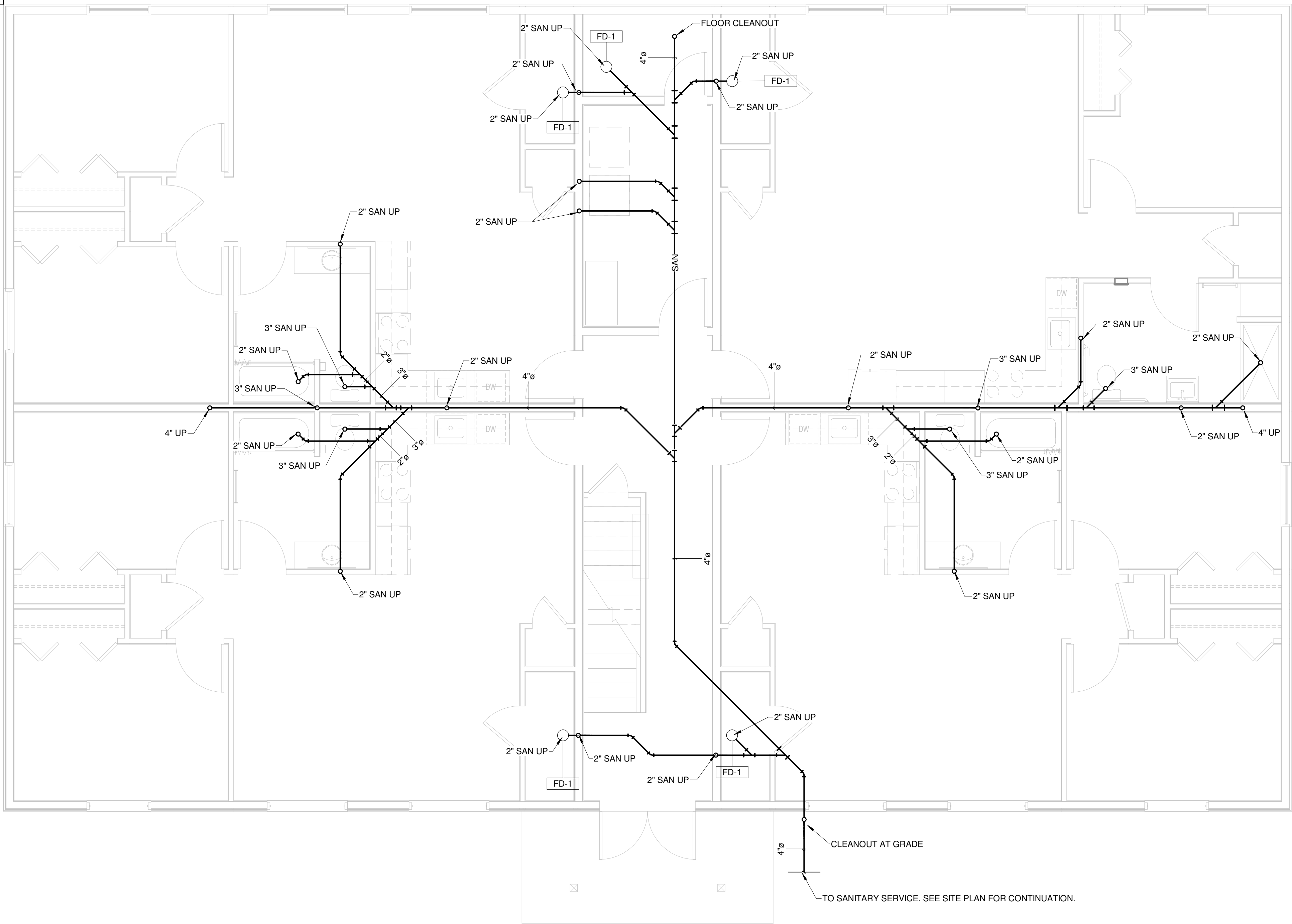
PLUMBING PIPE INSULATION SCHEDULE			
SYSTEM TYPE	INSUL. TYPE	PIPE SIZES	INSULATION THICKNESS
DOMESTIC HOT WATER SUPPLY	P-1	1-1/4" AND SMALLER 1-1/2" AND LARGER	1" 1-1/2"
DOMESTIC COLD WATER	P-1 OR P-5	1-1/4" AND SMALLER 1-1/2" AND LARGER	1/2" 1"

PIPING INSULATION SCHEDULE	
INSULATION TYPE	DESCRIPTION
P-1	ASTM C547, MOLDED GLASS FIBER PIPE INSULATION, OPERATING TEMPERATURE RANGE: 0 TO 850 DEG F, VAPOR BARRIER JACKET: ASTM C1136, TYPE I, FACTORY APPLIED REINFORCED FOIL DRAFT WITH SELF SEALING ADHESIVE JOINTS.
P-5	ASTM C534, TYPE I, FLEXIBLE, CLOSED CELL ELASTOMERIC INSULATION, TUBULAR, OPERATING TEMPERATURE RANGE: MINUS 70 TO 180 DEG F.

NOTE: ALL PLUMBING FIXTURES AND WATER HEATERS NOT LISTED BELOW ARE SUPPLIED BY OWNER, INSTALLED BY PLUMBING CONTRACTOR (PC). PC SHALL PROVIDE ALL SUPPLY LINES, TRAPS, TRAP WRAPS, AND THERMOSTATIC MIXING VALVES.

PLUMBING FIXTURE SCHEDULE												
ID	ADA	DESCRIPTION	MANUFACTURER	MODEL NUMBER	MATERIAL	BASIN DIMENSIONS (IN)	TRIM MANUF.	TRIM MODEL	HOT WATER (IN)	COLD WATER (IN)	SANITARY (IN)	NOTES
FD-1	-	6" DIAMETER FLOOR DRAIN WITH STRAINER	ZURN	Z415B	CAST IRON	-	-	-	-	-	2"	1
WB-1	-	WASHING MACHINE OUTLET BOX WITH ARRESTORS	SIOUX CHIEF	696	PLASTIC	-	-	-	1/2"	1/2"	2"	

1. PROVIDE WITH TRAP SEAL EQUAL TO "QUAD-CLOSE" BY J.R. SMITH.



A2 UNDERSLAB PLUMBING PLAN
1/4" = 1'-0"

IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES
BESSEMER, MI

DATE:
02/07/2024

ISSUED FOR:
BIDSET

IRON BELLE APARTMENTS
BUCKNELL ENTERPRISES
PROJECT NO: B280-03239

DESIGNED BY: SP
DRAWN BY: SP
CHECKED: GP
APPROVED: GP

UNDERSLAB
PLUMBING PLAN

P100



www.upea.com

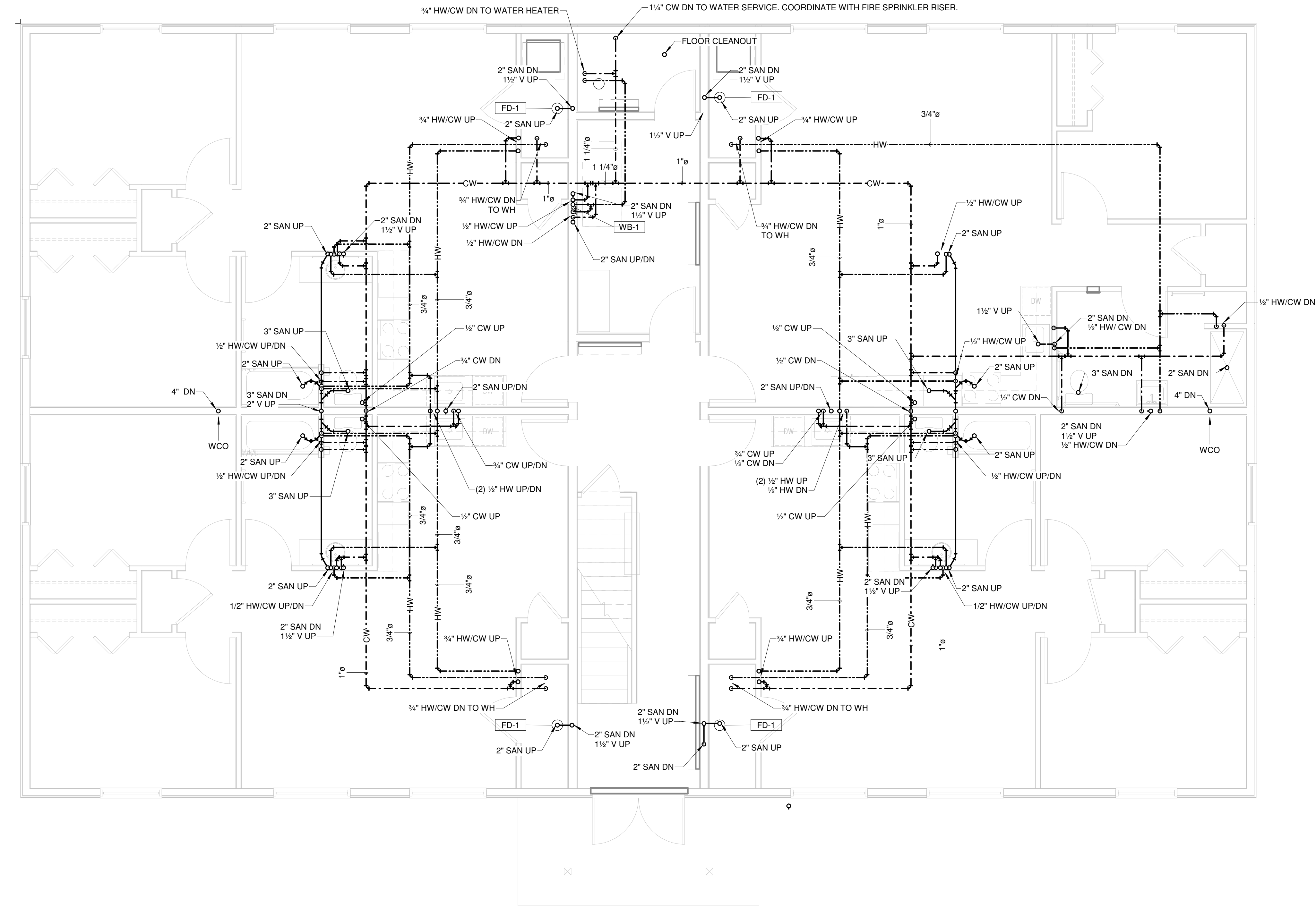


D

C

B

A



A2 FIRST FLOOR PLUMBING PLAN
1/4" = 1'-0"



IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES
BESSEMER, MI

DATE:

02/07/2024

ISSUED FOR:

BIDSET

IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES

PROJECT NO: B280-03239

DESIGNED BY: SP

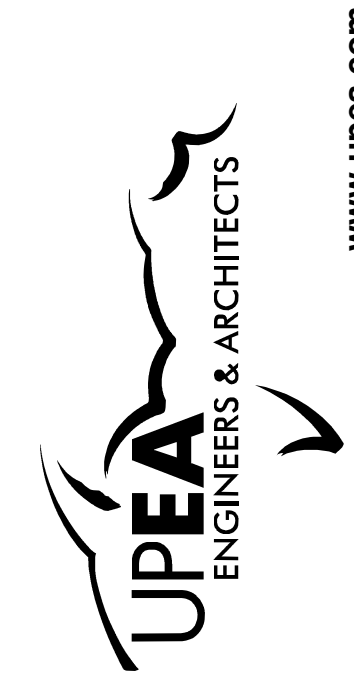
DRAWN BY: SP

CHECKED: GP

APPROVED: GP

FIRST FLOOR
PLUMBING PLAN

P101





P102

IRON BELLE APARTMENTS

**BUCKNELL ENTERPRISES
BESSEMER, MI**



1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218
219
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269
270
271
272
273
274
275
276
277
278
279
280
281
282
283
284
285
286
287
288
289
290
291
292
293
294
295
296
297
298
299
300
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
327
328
329
330
331
332
333
334
335
336
337
338
339
340
341
342
343
344
345
346
347
348
349
350
351
352
353
354
355
356
357
358
359
360
361
362
363
364
365
366
367
368
369
370
371
372
373
374
375
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
84

DATE:	02/07/2021
-------	------------

02/07/2024

BIDSET

BIDSET

BUICKNEI | ENTERPRISES

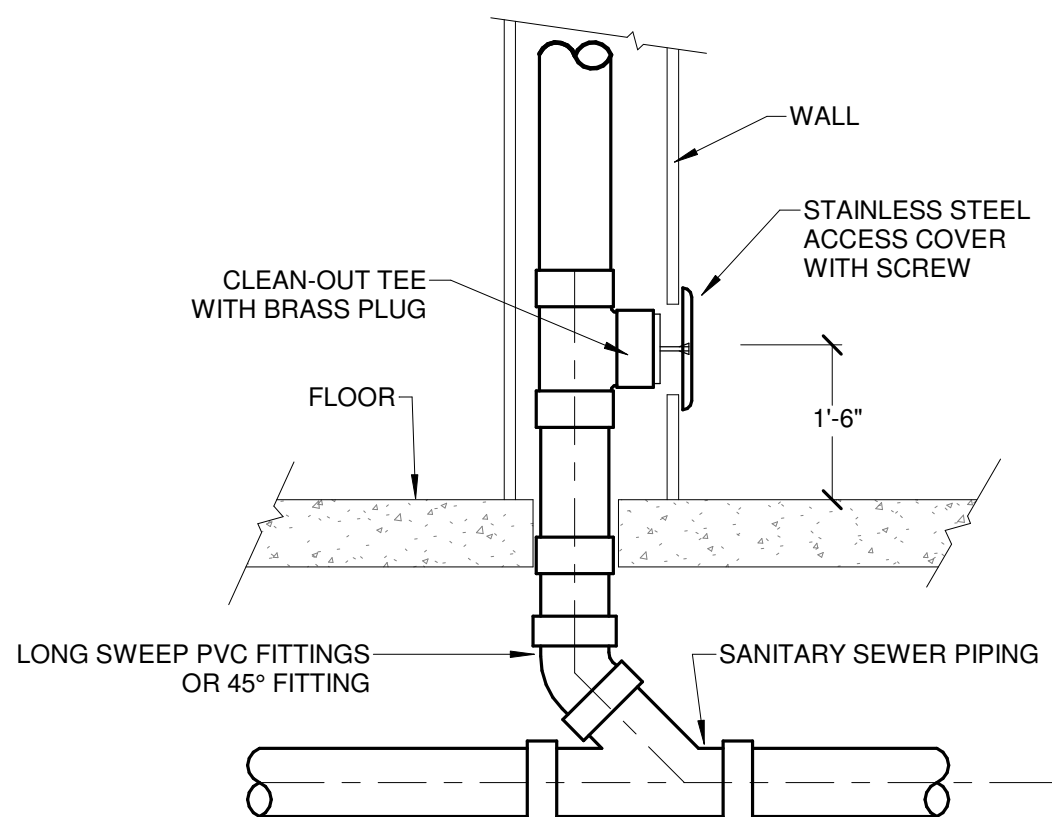
PROJECT NO: B280-03239

DESIGNED BY: SP
DRAWN BY: SP

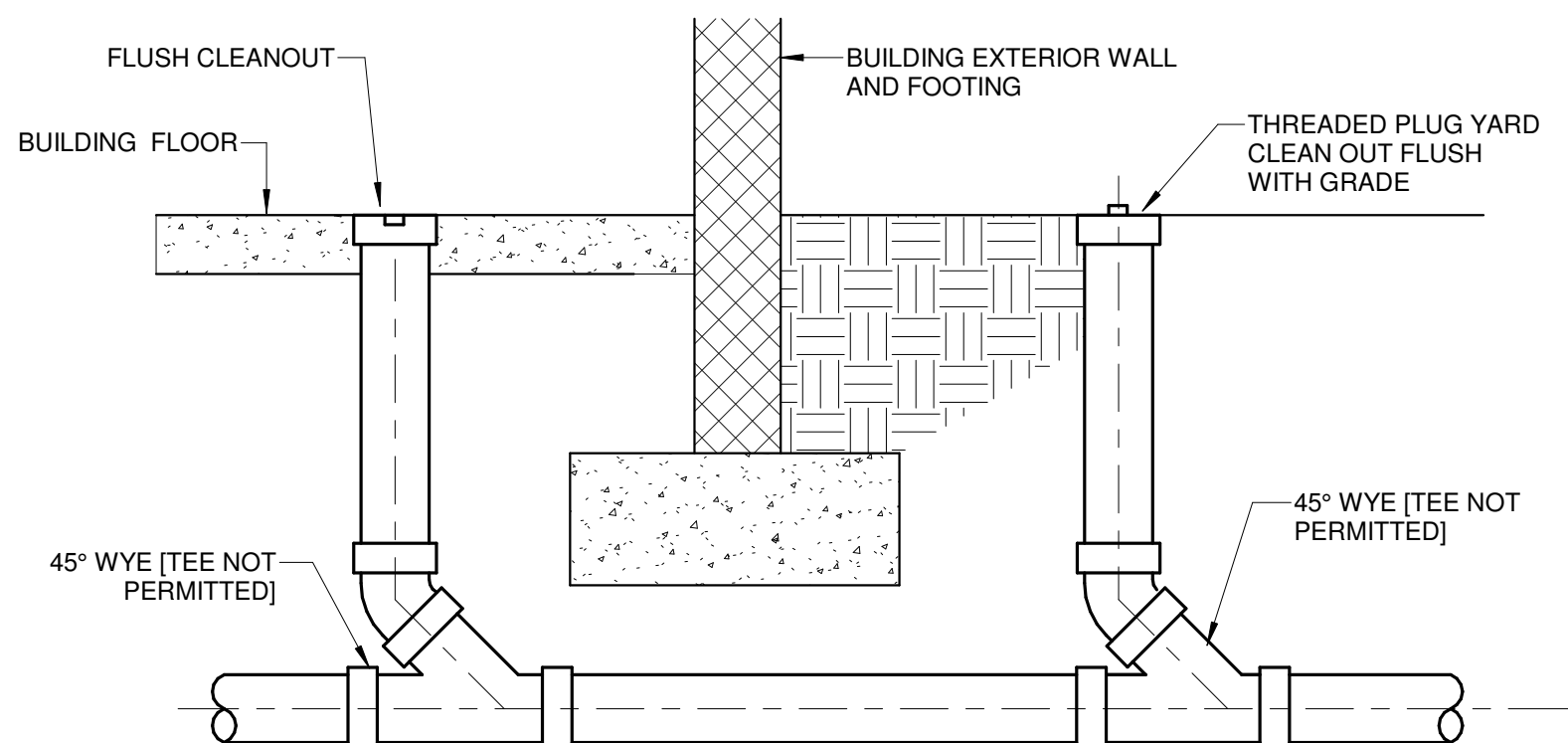
DESIGNED BY: SP
DRAWN BY: SP

DESIGNED BY: SP
DRAWN BY: SP

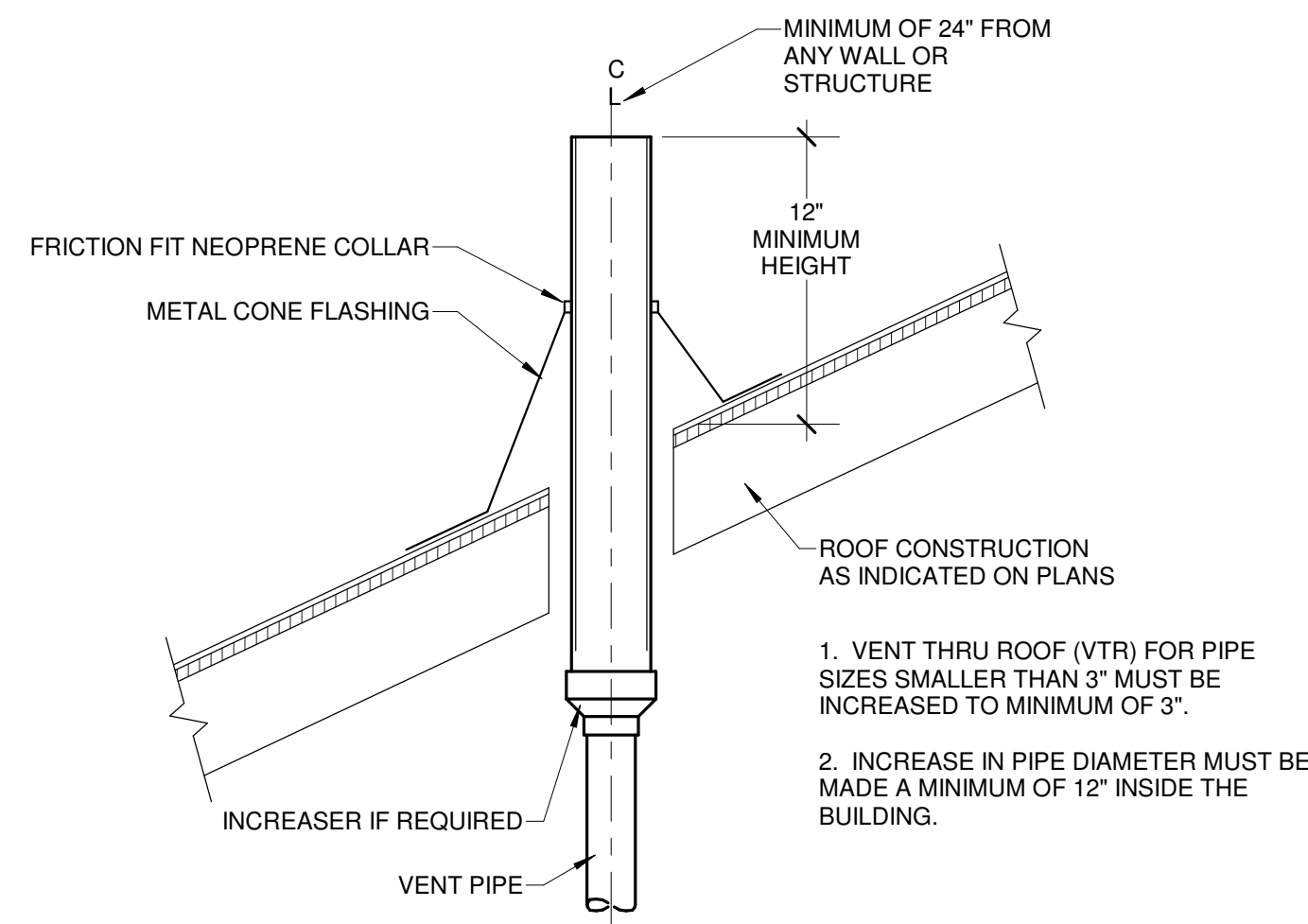
P-1B280-03239 Bucknell Enterprise-Iron Belle Apartments - Phase II ESA Dugs/Mechanical EC280-03239 Bucknell Enterprise-Iron Belle Apartments_Mechanical I



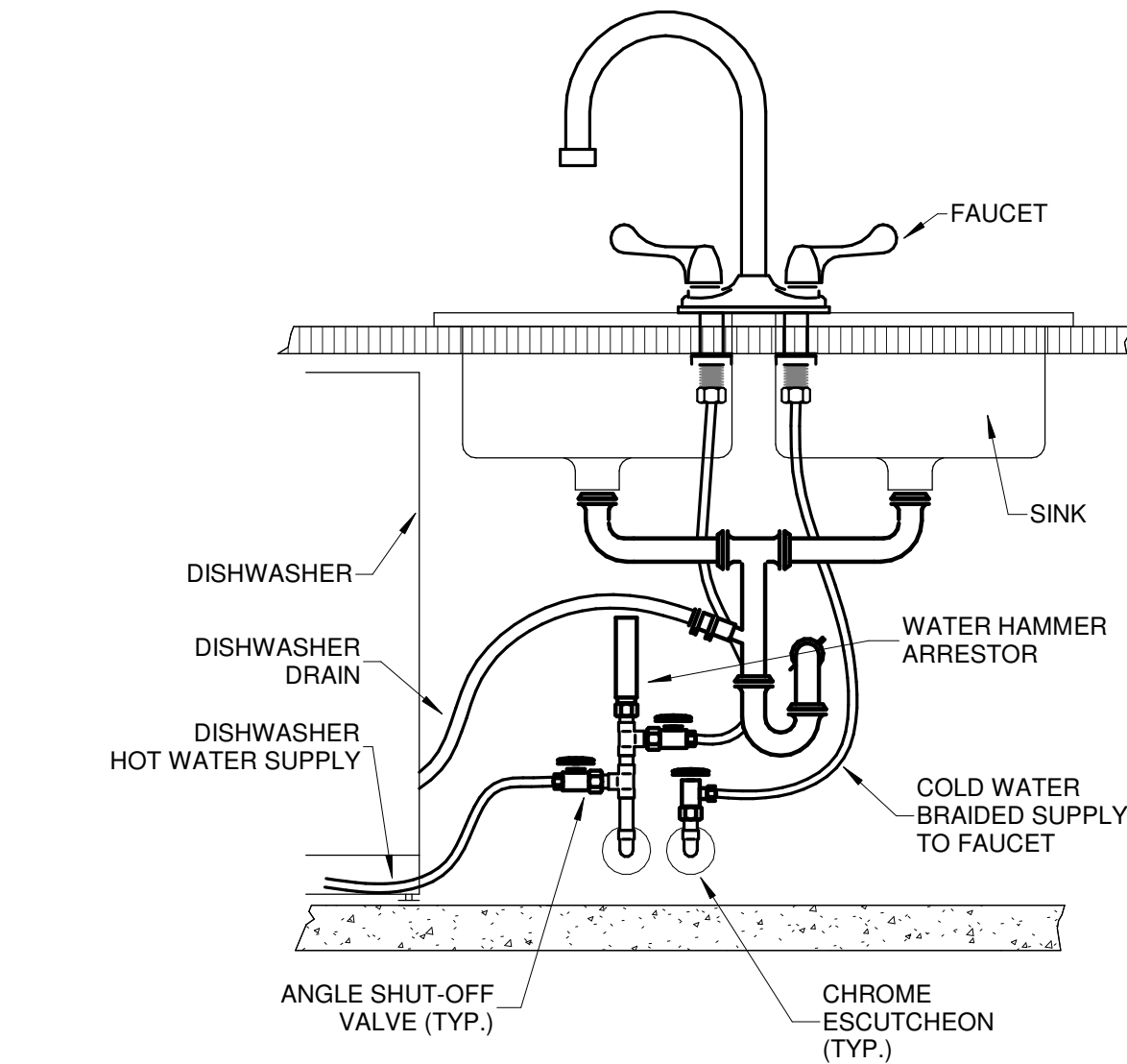
A1 WALL CLEANOUT DETAIL
NOT TO SCALE



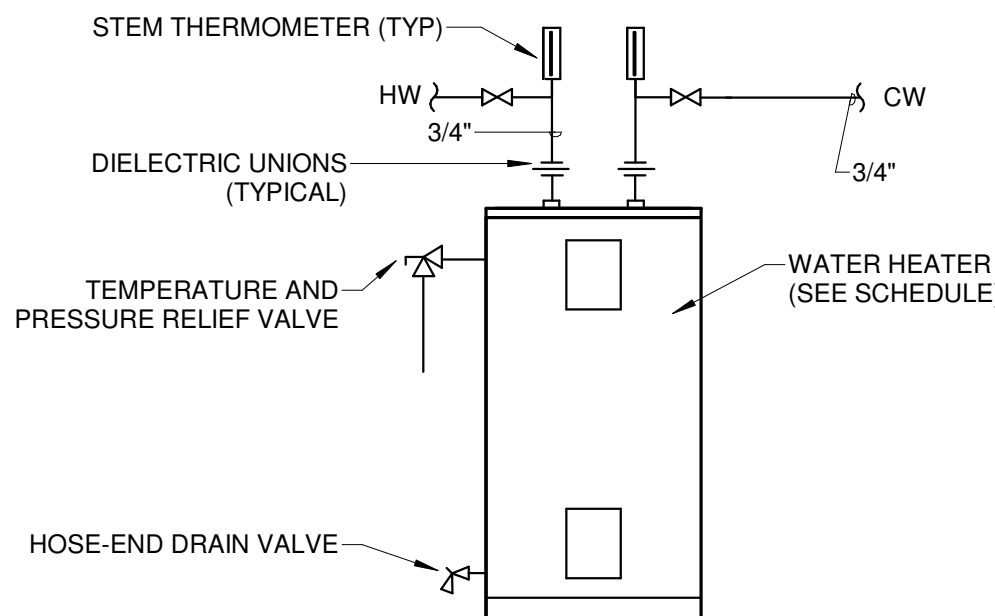
A2 SERVICE CLEANOUT DETAIL
NOT TO SCALE



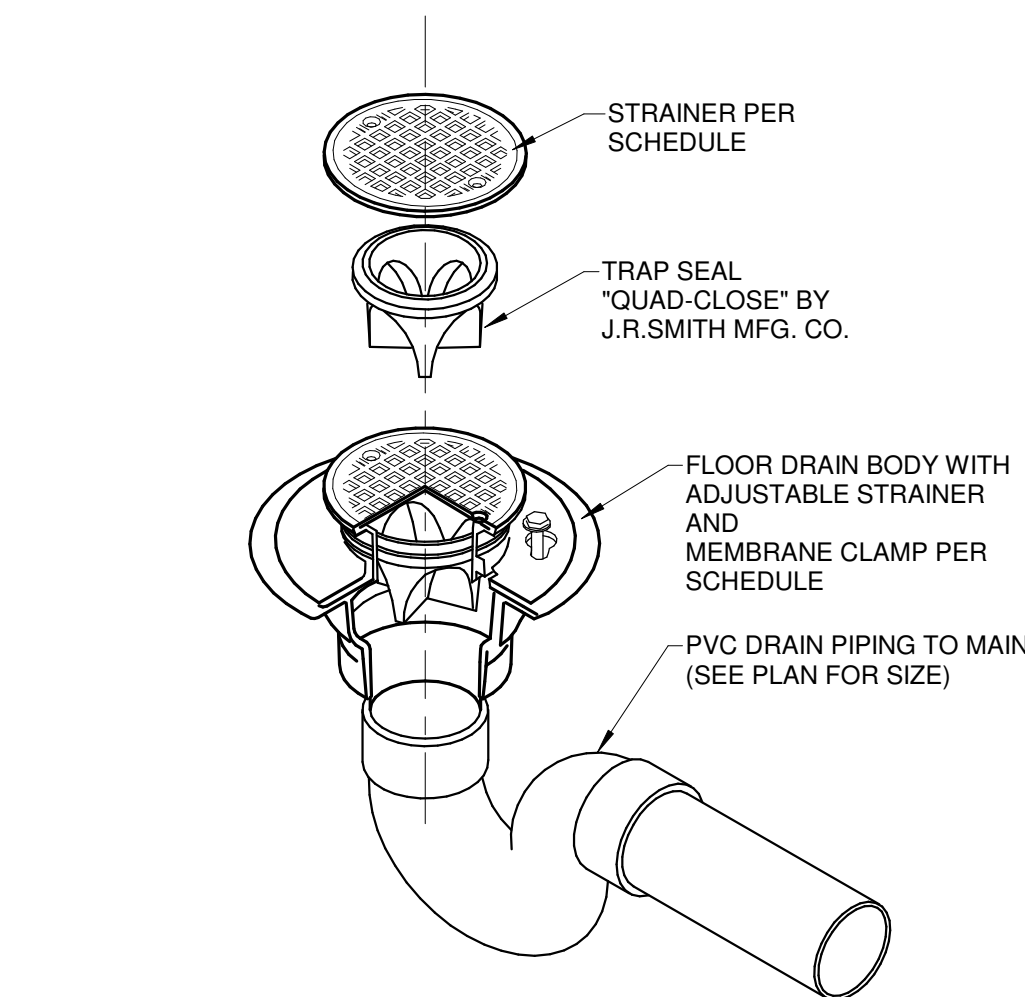
A4 VENT THROUGH ROOF DETAIL
NOT TO SCALE



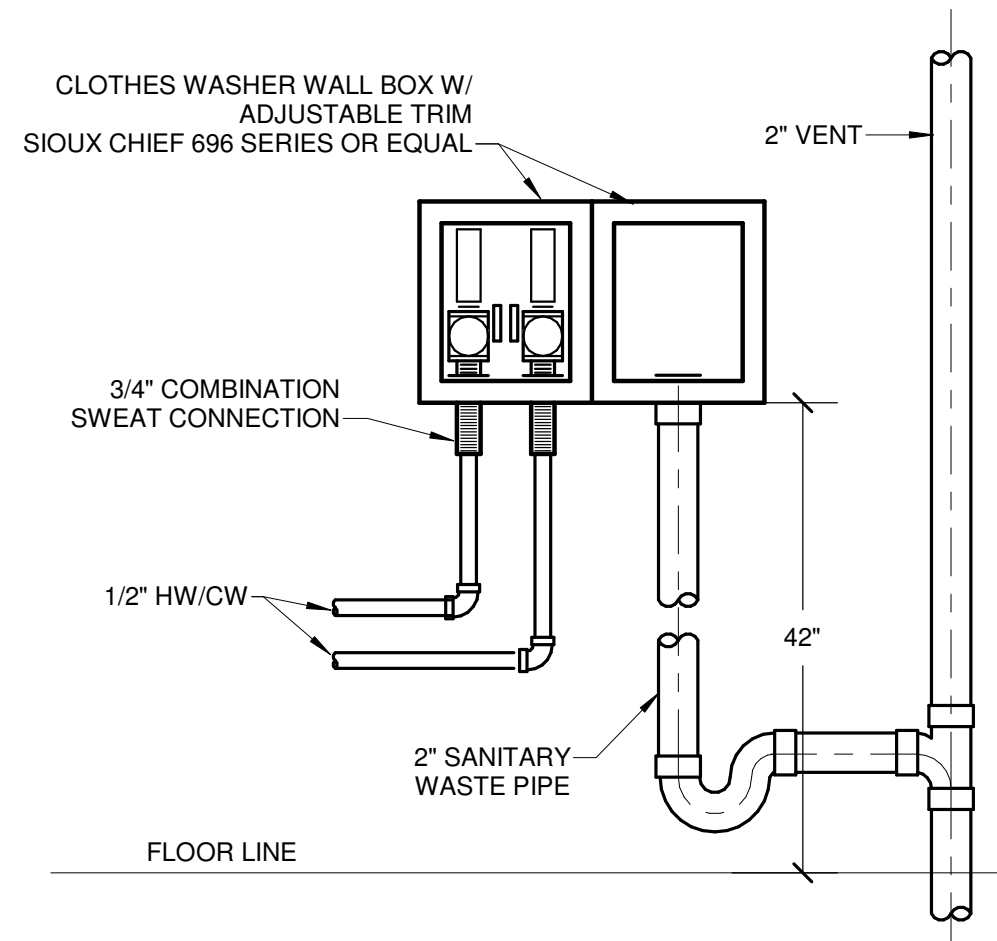
C3 DISHWASHER KITCHEN SINK DETAIL
NOT TO SCALE



B3 WATER HEATER PIPING DIAGRAM
NOT TO SCALE



C5 FLOOR DRAIN W/TRAP SEAL
FLOOR DRAIN DETAIL
NOT TO SCALE



B4 WASHING MACHINE OUTLET DETAIL
NOT TO SCALE

D
C
B
A

1

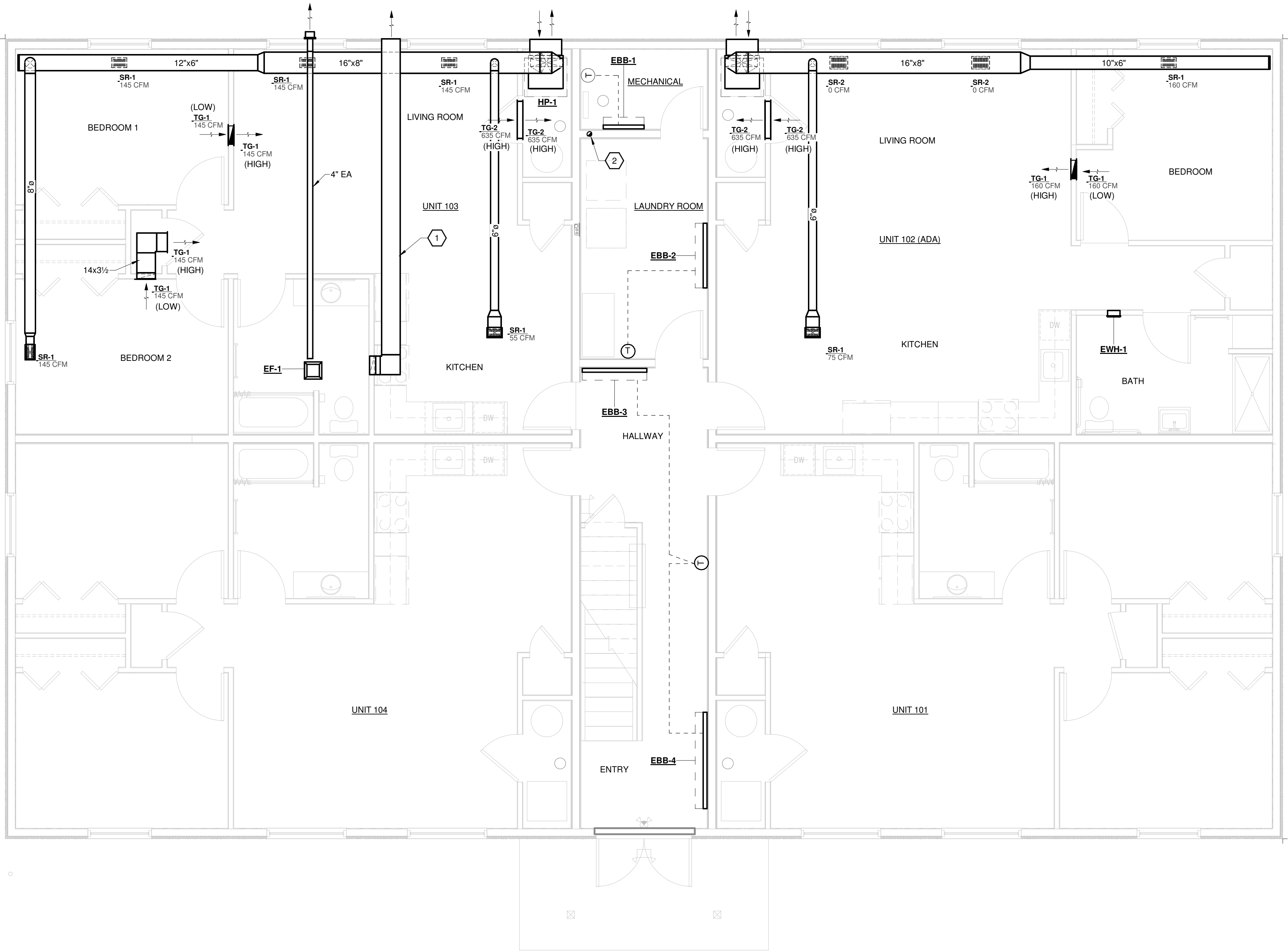
2

3

4

5

- GENERAL NOTES:**
- UNIT 103 MECHANICAL LAYOUT IS TYPICAL OF ALL UNITS EXCEPT ADA UNIT 102.
 - TRANSFER GRILLE LOCATIONS SHALL BE 6" FROM THE FLOOR IN THE BEDROOMS AND 6" FROM THE CEILING IN ALL OTHER AREAS. HIGH GRILLES MAY NEED TO BE FIELD ADJUSTED TO FIT ABOVE DOORS.
 - SUPPLY DUCTS ON FIRST FLOOR DO NOT NEED INSULATION.
 - HP-1 IS TYPICAL OF ALL UNITS.
- KEYNOTES**
- PROVIDE EXHAUST DUCT AND END CAP FROM OWNER PROVIDED HOOD. COORDINATE WITH OWNER AND GC.
 - PROVIDE DRYER BOX AND EXHAUST DUCT TO WALL CAP. TERMINATE A MINIMUM OF 3' ABOVE INTAKE AIR FOR HEAT PUMPS IN UNITS ABOVE.



A1 FIRST FLOOR MECHANICAL PLAN
1/4" = 1'-0"



IRON BELLE APARTMENTS
BUCKNELL ENTERPRISES
BESSEMER, MI

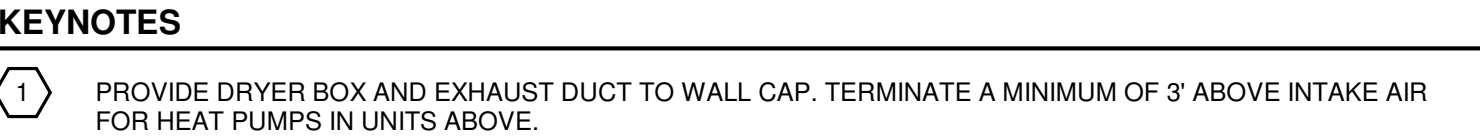
ISSUED FOR:	DATE:	
	BIDSET	02/07/2024
IRON BELLE APARTMENTS BUCKNELL ENTERPRISES PROJECT NO: B280-03239	DESIGNED BY:	SP
	DRAWN BY:	SP
	CHECKED:	GP
	APPROVED:	GP

**FIRST FLOOR
MECHANICAL PLAN**

M101



www.upea.com



**BUCKNELL ENTERPRISES
BESSEMER, MI**



www.upea.com

[illegible]

IRON BELLE APARTMENTS

PROJECT NO:	B280-03239
DESIGNED BY:	SP
DRAWN BY:	SP
CHECKED:	GP
APPROVED:	GP

SECOND FLOOR MECHANICAL PLAN

M102

2/7/2024 9:00:26 AM **A** P:\B280\03230 Buckwell Entertece\Iron Belle Apartments - Phase II F&D\Civil\Mechanical\B280_03230 Buckwell Entertece\Iron Belle Apartments Mechanical of

VENTILATION CALCULATION SCHEDULE - R-2, R-3 OR R-4 OCCUPANCIES, THREE STORIES OR LESS							
ROOM NAME/NUMBER	FLOOR AREA, A (SQ FT), or QTY.	BEDROOMS, N _{br} (QTY)	MINIMUM VENTILATION RATES (2015 MMC 403.3.2)				
			OUTDOOR AIRFLOW RATE, (CFM); Q _{oa} (0.01 A _{floor} + 7.5 (Nbr +1)	OUTDOOR AIRFLOW RATE, (CFM); @ 0.6 CFM/SQ FT	EXHAUST RATE, EACH (CFM)	OUTDOOR AIRFLOW RATE - TOTAL (CFM)	EXHAUST AIRFLOW RATE - TOTAL (CFM)
2-BEDROOM UNITS							
Floor Area/Quantity of Bedrooms	843	2	30.93			31	
KITCHENS - Int Vent (PER ROOM)	1				100		100
TOILET/BATHROOM - Int Vent (PER ROOM)	1				50		50
TOTAL						31	150
1 BEDROOM UNIT							
Floor Area/Quantity of Bedrooms	843	1	23.43			24	
KITCHENS - Int Vent (PER ROOM)	1				100		100
TOILET/BATHROOM - Int Vent (PER ROOM)	1				50		50
TOTAL						24	150

DUCTWORK INSULATION SCHEDULE		
SYSTEM TYPE	INSULATION TYPE	INSULATION OR LINER THICKNESS
SUPPLY DUCTS (EXTERNALLY INSULATED)	D-1, D-2, D-6	1-1/2"
EXHAUST DUCTS WITHIN 10 FEET OF EXTERIOR OPENINGS	D-1, D-2	1"

DUCTWORK INSULATION TYPE SCHEDULE	
INSULATION TYPE	DESCRIPTION
D-1	ASTM C1290, TYPE III, FLEXIBLE GLASS FIBER, COMMERCIAL GRADE WITH FACTORY APPLIED REINFORCED FOIL JACKET MEETING ASTM C1136, TYPE II, MAX. OPERATING TEMPERATURE: 250 DEG. F, DENSITY: 1.5 LBS/CU FT.
D-2	ASTM C612, TYPE IA OR IB, RIGID GLASS FIBER, WITH FACTORY APPLIED ALL SERVICE FACING MEETING ASTM C1136, TYPE II, DENSITY: 6.0 LBS/CU FT.
D-6	ASTM C534, TYPE II, FLEXIBLE, CLOSED CELL ELASTOMERIC INSULATION, SHEET, SERVICE TEMPERATURE RANGE: MINUS 58 TO 180 DEG F.

ELECTRIC WALL HEATER SCHEDULE						
ID	MANUFACTURER	MODEL	VOLTAGE	PHASE	WATTS	AMPS
EW-H-1	QMARK	CZ15112T	120	1	750	6.3

COORDINATE INSTALLATION WITH ELECTRICAL CONTRACTOR.
PROVIDE WITH LINE VOLTAGE THERMOSTAT LOCATED ABOVE HEATER. MOUNT THERMOSTAT PER ADA REQUIREMENTS.

ELECTRIC BASEBOARD HEATER SCHEDULE							
ID	MANUFACTURER	MODEL	WATTS	LENGTH	VOLTAGE	PHASE	AMPS
EBB-1	QMARK	25426	500	2'-6"	240	1	2.1
EBB-2	QMARK	2544	1000	4'-0"	240	1	4.2
EBB-3	QMARK	2544	1000	4'-0"	240	1	4.2
EBB-4	QMARK	2546	1500	6'-0"	240	1	6.3
EBB-5	QMARK	25426	500	2'-6"	240	1	2.1
EBB-6	QMARK	2544	1000	4'-0"	240	1	4.2
EBB-7	QMARK	2544	1000	4'-0"	240	1	4.2
EBB-8	QMARK	2544	1000	4'-0"	240	1	4.2

COORDINATE INSTALLATION WITH ELECTRICAL CONTRACTOR.
INSTALL PER MANUFACTURER'S REQUIREMENTS. MAINTAIN ALL RECOMMENDED CLEARANCES.
PROVIDE WITH LINE VOLTAGE THERMOSTAT AND LOCKABLE COVER.

DIFFUSER, GRILLE, REGISTER SCHEDULE								
ID	DESCRIPTION	MANUFACTURER	MODEL NUMBER	MOUNTING FRAME	NECK SIZE (IN)	THROW	MATERIAL	Notes
SR-1	SUPPLY REGISTER	HART AND COOLEY	631	SURFACE	10 x 6	3-WAY	STEEL	1,2,3
SR-2	SUPPLY REGISTER	HART AND COOLEY	631	SURFACE	12 x 8	3-WAY	STEEL	1,2,3
TG-1	TRANSFER GRILLE	HART AND COOLEY	672/A672	SURFACE	14 x 6	-	STEEL/ALUMINUM	1,2
TG-2	TRANSFER GRILLE	HART AND COOLEY	672/A672	SURFACE	30 x 8	-	STEEL/ALUMINUM	1,2

- BORDER TYPE SHALL BE COMPATIBLE WITH ARCHITECTURAL CEILING TYPE IN WHICH THE DEVICE IS LOCATED.
- SEE PLAN FOR LOCATIONS AND QUANTITIES OF AIR DEVICES.
- PROVIDE WITH OPPOSED BLADE DAMPER ADJUSTABLE FROM FACE.

VARIABLE REFRIGERANT PACKAGED HEAT PUMP											
ID	MANUFACTURER	MODEL NUMBER	AIRFLOW/BLOWER DATA		HEATING CAPACITY (MBH)	COOLING CAPACITY (MBH)	ELECTRIC COIL CAPACITY (WATTS)	ELECTRICAL			
			CFM	ESP (IN)				VOLTAGE	PHASE	AMPS	MOCP

COORDINATE ELECTRICAL CONNECTION WITH ELECTRICAL CONTRACTOR.
INSTALL PER MANUFACTURER'S INSTALLATION INSTRUCTIONS.
PROVIDE MINIMUM OUTDOOR AIR PER VENTILATION SCHEDULE.
PROVIDE WITH 7.5 KW ELECTRIC BACKUP COILS.
PROVIDE WITH LOUVER, DAMPER, DUCT AND ALL NECESSARY COMPONENTS FOR A COMPLETE INSTALLATION.
PROVIDE WITH PROGRAMMABLE THERMOSTAT AND WIRING.

FAN SCHEDULE												
ID	DESCRIPTION	MANUFACTURER	MODEL NUMBER	AIR FLOW (CFM)	EXT. SP. (FT)	RPM	DUCT CONNECTION (IN)	ELECTRICAL			Notes	
								VOLTS	PHASE	WATTS		
EF-1	BATHROOM EXHAUST FAN W/ LIGHT	PANASONIC	FV-0511VKL2	50-80-110	0.25	1121	4"Ø	120	1	10.5	1,2,3	

- COORDINATE ELECTRICAL REQUIREMENTS WITH ELECTRICAL CONTRACTOR.
- PROVIDE WITH WALL CAP AND INSULATED EXHAUST DUCTWORK PER DETAILS AND SCHEDULE.
- UNIT SHALL TURN ON WITH LOCAL SWITCH.



www.upea.com

IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES
BESSEMER, MI

DATE:

02/07/2024

ISSUED FOR:

BIDSET

IRON BELLE APARTMENTS

BUCKNELL ENTERPRISES
PROJECT NO: B280-03239

DESIGNED BY: SP
DRAWN BY: SP
CHECKED: GP
APPROVED: GP

MECHANICAL
SCHEDULES

M601